



Shore

Student Number:

Set:

Year 12
Mathematics Extension 1
Trial HSC Examination
2011

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided on the back page of this question paper
- All necessary working should be shown in every question

Note: Any time you have remaining should be spent revising your answers.

Total marks – 84

- Attempt Questions 1 – 7
- All questions are of equal value
- Start each question in a new writing booklet
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Total Marks – 84
Attempt Questions 1 – 7
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks)

Marks

- | | |
|---|----------|
| (a) The interval AB , where $A(1, -2)$ and $B(6, 7)$, is divided externally in the ratio $2 : 3$ by the point $P(x, y)$. Find the values of x and y . | 2 |
| (b) Find the domain and range of the function $f(x) = 2\sin^{-1} 3x$. | 2 |
| (c) Solve $x - 1 \leq \frac{2}{x}$. | 3 |
| (d) Differentiate $\sin^{-1}(\cos x)$. Write your answer in simplest form. | 2 |
| (e) Use the substitution $u = 1 + x$ to find $\int \frac{x}{\sqrt{1+x}} dx$. | 3 |

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

Question 2 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) The acceleration of a particle is given by $\ddot{x} = 4(x+1) \text{ ms}^{-2}$. Initially, the particle is at the origin and velocity is 2 ms^{-1} .

(i) Show that the velocity, v , at any position, x , is given by $v = 2x + 2$.

2

(ii) Hence show that $x = e^{2t} - 1$.

2

(b)

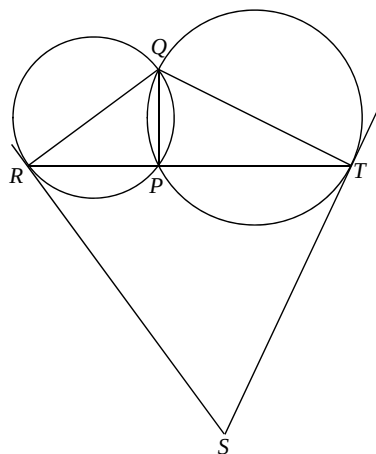
(i) Express $\cos \theta - \sqrt{3} \sin \theta$ in the form $R \cos(\theta + \alpha)$ where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$.

2

(ii) Hence, or otherwise, solve $\cos \theta - \sqrt{3} \sin \theta = 1$ for $0 \leq \theta \leq 2\pi$.

2

(c)



NOT TO
SCALE

Two circles intersect at P and Q . RPT is a straight line where R is a point on the first circle and T is a point on the second circle. The tangent at R and the tangent at T meet at S .

Copy the diagram into your booklet.

Prove that $QRST$ is a cyclic quadrilateral.

4

Question 3 (12 marks) Use a SEPARATE writing booklet

Marks

(a)

(i) Show that the equation $e^{-x} = \sin 2x$ has a root lying between 1 and 2.

1

(ii) By taking 1.5 as a first approximation, use Newton's method once, to obtain a better approximation to this root. (Give your answer correct to 2 decimal places).

2

(b) Let T be the temperature in a room at time t and let A be the temperature of its surroundings. Newton's Law of Cooling states that the rate of change of temperature T is proportional to $(T - A)$ i.e. $\frac{dT}{dt} = k(T - A)$.

(i) Verify that $T = A + Be^{kt}$ is a solution to $\frac{dT}{dt} = k(T - A)$.

1

(ii) If the temperature of a substance in a room of constant temperature 6°C is noted to be 29°C and in 40 minutes to be 14°C , find the value of B and k .

2

(iii) Find how long it takes for the temperature of the substance to reach 9°C . Give your answer to the nearest minute.

2

(c) The polynomial $P(x) = 2x^3 + kx^2 + 3x - 4$ has roots α , β and γ .

(i) Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.

2

(ii) If one root is the reciprocal of the other, find the third root and hence find the value of k .

2

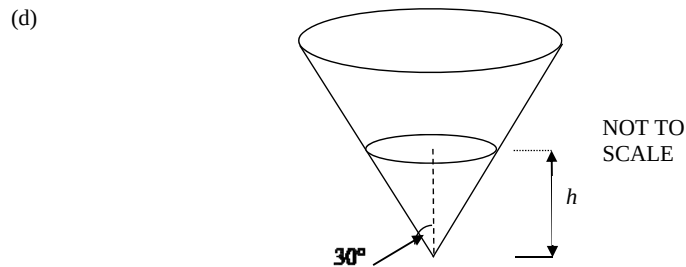
Question 4 (12 marks) Use a SEPARATE writing booklet

Marks

(a) Evaluate $\int_0^3 \frac{dx}{9+x^2}$. 2

(b) Find the term independent of x in the expansion $\left(x^2 + \frac{2}{x}\right)^9$. 3

(c) The region bounded by the curve $y = 3\sin 2x$, the x -axis and the line $x = \frac{\pi}{4}$ is rotated about the x -axis to form a solid of revolution.
Find the volume of the solid formed. 3



A hollow cone with vertical angle 60° is held with its axis vertical and vertex downwards.

Sand is being poured into the cone at a uniform rate of 15 cubic metres per second.

(i) Show that when the sand level has reached a height of h metres, the volume of the sand in the cone, in cubic metres, is given by $V = \frac{1}{9}\pi h^3$. 2

(ii) Find the rate at which the sand level is rising when its depth is 4 metres. (Express your answer in terms of π). 2

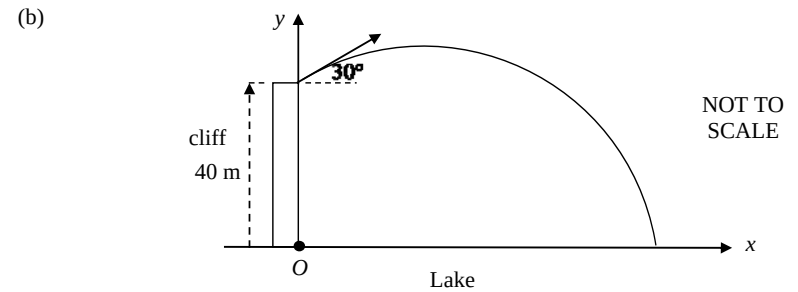
Question 5 (12 marks) Use a SEPARATE writing booklet

Marks

(a) Consider the function $f(x) = e^{x+2}$ 2

(i) Find the inverse function $f^{-1}(x)$. 2

(ii) Sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ clearly indicating any intercepts with the axes. 2



A pebble is projected from the top of a vertical cliff with velocity 20 ms^{-1} at an angle of elevation of 30° . The cliff is 40 metres high and overlooks a lake.

Assume that, t seconds after release of the pebble, the horizontal displacement from O is given by $x = 10\sqrt{3}t$. (DO NOT prove this).

(i) Assuming the acceleration due to gravity is 10 ms^{-2} , show that the pebble's vertical displacement is given by $y = -5t^2 + 10t + 40$. 2

(ii) Calculate the time which elapses before the pebble hits the lake and the distance of the point of impact from the foot of the cliff. 3

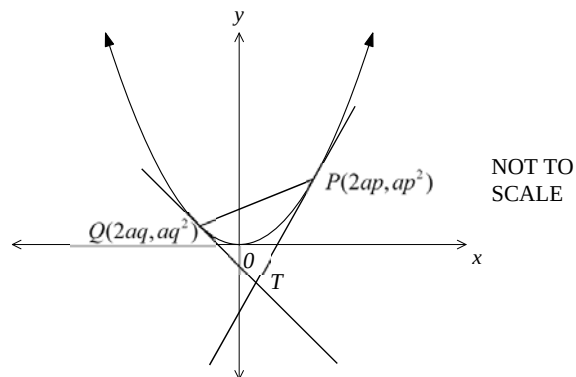
(iii) Find the angle and the speed at which the pebble hits the lake. 3

Question 6 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) Use the process of mathematical induction to prove that $2^{3n} - 1$ is divisible by 7 for all integers $n \geq 1$. 3

(b)



In the diagram above a focal chord PQ intersects the parabola $x^2 = 4ay$ at points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$. The tangents to the parabola at point P and point Q intersect at T .

- (i) Show that the equation of the chord PQ is given by $y = \left(\frac{p+q}{2}\right)x - apq$. 2
- (ii) Show that $pq = -1$. 1
- (iii) Show that the acute angle between the focal chord QP and the tangent TP to the parabola at P is given by $\tan^{-1}|q|$. 3

- (c) Consider the expansion of $\left(\frac{1}{3} + 2x\right)^{18}$.

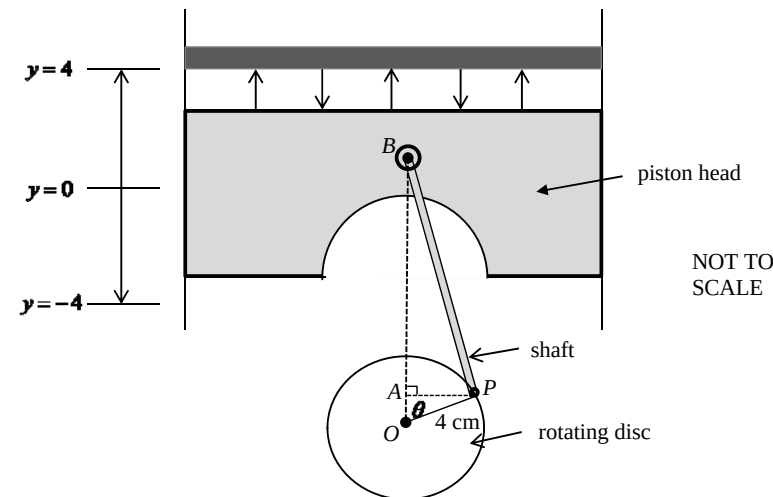
- (i) Show that the $(k+1)$ th term is given by $T_{k+1} = {}^{18}C_k \frac{2^k}{3^{18-k}} x^k$. 1

- (ii) Hence, or otherwise, find the greatest coefficient of the expansion $\left(\frac{1}{3} + 2x\right)^{18}$. 2

Question 7 (12 marks) Use a SEPARATE writing booklet

Marks

The diagram below illustrates the movement of a piston head in a combustion engine. As the disc rotates, the shaft (BP) moves the piston head up and down. The radius of the rotating disc is 4 cm and the vertical displacement of the piston at any time t seconds is given by the equation $y = 4\cos 10t$.



- (i) Find $\frac{dy}{dt}$ and $\frac{d^2y}{dt^2}$ and hence show that the piston moves in simple harmonic motion. 2
- (ii) State the period of the motion in exact form. 1
- (iii) Find the exact velocity of the piston when $y = 3$ cm at the first time it reaches this point. 2
- (iv) Show that $OB = 4\cos\theta + \sqrt{BP^2 - 16\sin^2\theta}$ 3

Question 7 continued over the page.....

- (v) If the disc is rotating at a rate of 10 radians/second, and

3

$$\frac{d(OB)}{dt} = \frac{d(OB)}{d\theta} \times \frac{d\theta}{dt} \text{ show that}$$

$$\frac{d(OB)}{dt} = -40 \sin \theta \left[1 + \frac{\cos \theta}{\sqrt{\left(\frac{BP}{4}\right)^2 - \sin^2 \theta}} \right]$$

- (vi) Calculate $\frac{d(OB)}{dt}$, the rate at which the distance OB is changing over time, when $\theta = \pi$.

1

End of Paper

2011 EXTENSION 1 TRIAL HSC

Question 1:

a) $(1, -2) \quad (6, 7)$
 $-2: 3$

$$x = \frac{mx_1 + nx_2}{m+n} \quad y = \frac{my_1 + ny_2}{m+n}$$

$$= \frac{-2 \times 6 + 3 \times 1}{1} \quad = \frac{-2 \times 7 + 3 \times 2}{1}$$

$$= -9 \quad = -20$$

$$\therefore P(-9, -20) \quad [2]$$

b) $f(x) = 2\sin^{-1} 3x$

Domain: $-1 \leq 3x \leq 1$
 $-\frac{1}{3} \leq x \leq \frac{1}{3}$

Range: $-\frac{\pi}{2} \leq \frac{y}{2} \leq \frac{\pi}{2}$
 $-\pi \leq y \leq \pi \quad [2]$

c) Critical points:

1. $x \neq 0$

2. Solve $x-1 = \frac{2}{x}$
 $x^2 - x = 2$

$$x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0$$

$$\therefore x = -1 \text{ or } 2$$



3. Test $x = -2$ ✓ $\therefore x \leq -1$ or $0 < x \leq 2$

$$x = -\frac{1}{2} \times$$

$$x = 1 \times$$

$$x = 3 \times$$

[3]

d) $y = \sin^{-1}(\cos x)$
 $\frac{dy}{dx} = \frac{1}{\sqrt{1-\cos^2 x}} \times -\sin x$
 $= \frac{-\sin x}{\sqrt{\sin^2 x}}$
 $= \frac{-\sin x}{\sin x}$
 $= -1 \quad [2]$

e) $u = 1+x \Rightarrow x = u-1$

$$\frac{du}{dx} = 1$$

$$du = dx$$

$$\int \frac{x}{\sqrt{1+x}} dx = \int \frac{u-1}{\sqrt{u}} du$$

$$= \int \frac{u-1}{u^{1/2}} du$$

$$= \int (u^{1/2} - u^{-1/2}) du$$

$$= \frac{2u^{3/2}}{3} - 2u^{1/2} + C$$

$$= \frac{2}{3} \sqrt{1+x}^3 - 2\sqrt{1+x} + C$$

[3]

Question 2:

a) $\ddot{x} = 4(x+1) \quad t=0, x=0 \quad v=2$

$$\int \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 4 \int (x+1) dx$$

$$\frac{1}{2} v^2 = 4 \left[\frac{x^2}{2} + x \right] + C$$

$$x=0 \quad v=2$$

$$2 = 4(0+0) + C$$

$$2 = C$$

$$\frac{1}{2} v^2 = 4 \left(\frac{x^2}{2} + x \right) + 2$$

$$v^2 = 8 \left(\frac{x^2}{2} + x \right) + 4$$

$$= 4x^2 + 8x + 4$$

$$= 4(x^2 + 2x + 1)$$

$$= 4(x+1)^2$$

$$v = \pm 2(x+1) \quad v > 0$$

$$\therefore v = 2(x+1) \text{ m/s} \quad [2]$$

(ii) $v = 2x+2$

$$\frac{dx}{dt} = 2x+2$$

$$\frac{dx}{2(x+1)} = \frac{1}{2} dt$$

$$t = \frac{1}{2} \int \frac{1}{x+1} dx$$

$$t = \frac{1}{2} \ln(x+1) + C$$

$$t=0 \quad x=0$$

$$0 = \frac{1}{2} \ln 1 + C$$

$$0 = C$$

$$\therefore t = \frac{1}{2} \ln(x+1)$$

$$2t = \ln(x+1)$$

$$x+1 = e^{2t}$$

$$x = e^{2t} - 1$$

Which is required [2]

(b) $\cos \theta - \sqrt{3} \sin \theta = R \cos(\theta + \alpha)$
 $R \cos \theta \cos \alpha - R \sin \theta \sin \alpha = R \cos(\theta + \alpha)$

$$\therefore R \cos \alpha = 1 \quad R \sin \alpha = \sqrt{3}$$

$$\cos \alpha = \frac{1}{R} \quad \sin \alpha = \frac{\sqrt{3}}{R}$$

$$\therefore R^2 = 3+1$$

$$R = 2$$

$$\tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3}$$

$$\therefore \cos \theta - \sqrt{3} \sin \theta = 2 \cos \left(\theta + \frac{\pi}{3} \right) \quad [2]$$

(ii) $2 \cos \left(\theta + \frac{\pi}{3} \right) = 1$

$$\cos \left(\theta + \frac{\pi}{3} \right) = \frac{1}{2}$$



$$\left[\begin{array}{l} \text{for } 0 \leq \theta \leq 2\pi \\ \frac{\pi}{3} \leq \theta + \frac{\pi}{3} \leq \frac{7\pi}{3} \end{array} \right]$$

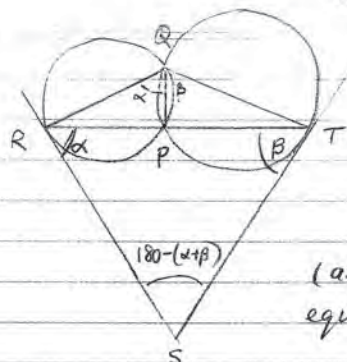
$$\therefore \theta + \frac{\pi}{3} = \frac{\pi}{3}, \frac{7\pi}{3} \text{ or } \theta + \frac{\pi}{3} = \frac{5\pi}{3}$$

$$\theta = 0, 2\pi, \text{ or } \frac{4\pi}{3}$$

$$\text{ie } \theta = 0, \frac{4\pi}{3} \text{ or } 2\pi \quad [2]$$

-3-

(c)



$\angle SRT = \angle RQP = \alpha$
(angle between a chord & a tangent is equal to the $\angle$ in the alternate segment)

$\angle STP = \angle TQP = \beta$
(angle between a chord & a tangent is equal to the $\angle$ in the alternate segment)

Now $\angle RQT = \alpha + \beta$ (by addition)

In $\triangle SRT$, $\angle RST = 180 - (\alpha + \beta)$ (Angle sum of $\triangle RTS$)

$$\angle RQT + \angle RST = (\alpha + \beta) + 180 - (\alpha + \beta) = 180^\circ$$

$\therefore \angle RQT$ and $\angle RST$ are supplementary $\angle$ s.
Hence $RQTS$ is a cyclic quadrilateral. [4]

Question 3:

2)	$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$
(i) let $f(x) = e^{-x} - \sin 2x$	
$f(1) = e^{-1} - \sin 2 = -0.54...$	$= 1.5 - 0.082$
$f(2) = e^{-2} - \sin 4 = 0.89...$	$= 1.7568$
	$= 1.4533...$
	$= \underline{1.45} \quad (2 \text{ dp})$ [2]

Since $f(1) < 0$ and $f(2) > 0$ then there is a root between 1 and 2.

[1]

(ii) $f'(x) = -e^{-x} - 2\cos 2x$
 $f'(1.5) = -e^{-1.5} - 2\cos 3$
 $= 1.7568$
 $f(1.5) = e^{-1.5} - \sin 3$
 $= 0.082$

-4-

b) (i) $T = A + Be^{kt}$ (1)
 $\frac{dT}{dt} = Bke^{kt}$ (2)
 but from (1) $T - A = Be^{kt}$

$\therefore$ sub into (2)

$$\frac{dT}{dt} = k(T - A) \quad \therefore \text{a soln}$$

[1]

(ii) $t = 0 \quad T = 29 \quad t = 40 \quad T = 14^\circ$
 $A = 6^\circ$

When $t = 0, T = 29$
 $29 = 6 + Be^0$
 $23 = B$

$$T = 6 + 23e^{kt}$$

$$14 = 6 + 23e^{40k}$$

$$\frac{8}{23} = e^{40k}$$

$$\ln\left(\frac{8}{23}\right) = 40k$$

$$\frac{1}{40} \ln\left(\frac{8}{23}\right) = k$$

[2]

(iii) $t = ? \quad T = 9$

$$9 = 6 + 23e^{kt}$$

$$3 = 23e^{kt}$$

$$\frac{3}{23} = e^{kt}$$

$$\ln\left(\frac{3}{23}\right) = kt$$

$$\frac{1}{k} \ln\left(\frac{3}{23}\right) = t$$

$$t = 77.15 \quad [2]$$

$$t = \underline{77 \text{ minutes}} \quad (\text{nearest minute}) \quad \therefore \text{Term is } {}^9C_6 \cdot 2^6 = \underline{5376} \quad [3]$$

(c) $P(x) = 2x^3 + kx^2 + 3x - 4$

$$(1) \quad \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{xy + xz + yz}{xyz}$$

$$(2) \quad \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{xy + xz + yz}{xyz}$$

$$= \frac{3}{2}$$

$$= \frac{3}{4}$$

[2]

(ii) Let the roots be $\alpha, \frac{1}{2}, \beta$
 $\alpha \times \frac{1}{2} \times \beta = 2 \Rightarrow \beta = 2$

$$P(2) = 16 + 4k + 6 - 4 = 0$$

$$4k = -18$$

$$k = -4\frac{1}{2} \quad [2]$$

Question 4:

a) $\int_0^3 \frac{dx}{9+x^2} = \frac{1}{3} \left[\tan^{-1} \frac{x}{3} \right]_0^3$
 $= \frac{1}{3} [\tan^{-1} 1 - \tan^{-1} 0]$
 $= \frac{1}{3} \left(\frac{\pi}{4} - 0 \right)$
 $= \frac{\pi}{12} \quad [2]$

b) ${}^9C_k (x^2)^{9-k} (2x^{-1})^k$
 $= {}^9C_k 2^k \cdot x^{18-2k} \cdot x^{-k}$
 $= {}^9C_k 2^k \cdot x^{18-3k}$

Term independent of x is x^0
 $\therefore x^0 = x^{18-3k}$
 $0 = 18 - 3k$
 $k = 6$

$$c) V = \pi \int_0^{\pi/4} (9 \sin^2 2x) dx$$

$$= 9\pi \int_0^{\pi/4} \sin^2 2x dx$$

$$\left. \begin{aligned} \cos 2x &= \cos^2 x - \sin^2 x \\ &= 1 - 2\sin^2 x \\ 2\sin^2 x &= 1 - \cos 2x \\ \sin^2 x &= \frac{1}{2}(1 - \cos 2x) \\ \sin^2 2x &= \frac{1}{2}(1 - \cos 4x) \end{aligned} \right\}$$

$$\therefore V = \frac{9\pi}{2} \int_0^{\pi/4} (1 - \cos 4x) dx$$

$$= \frac{9\pi}{2} \left[x - \frac{1}{4} \sin 4x \right]_0^{\pi/4}$$

$$= \frac{9\pi}{2} \left[\left(\frac{\pi}{4} - \frac{1}{4} \sin \pi \right) \right.$$

$$\left. - \left(0 - \frac{1}{4} \sin 0 \right) \right]$$

$$= \frac{9\pi}{2} \left[\frac{\pi}{4} - 0 \right]$$

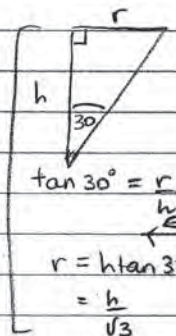
$$= \frac{9\pi^2}{8} \text{ units}^3 \quad [3]$$

$$d) V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi \left(\frac{h}{\sqrt{3}} \right)^2 h$$

$$= \frac{1}{3} \pi \frac{h^2}{3} h$$

$$= \frac{1}{9} \pi h^3$$



[2]

$$(ii) \text{ given } \frac{dv}{dt} = 15$$

$$V = \frac{1}{3} \pi h^3$$

$$\frac{dv}{dh} = \frac{1}{3} \pi h^2$$

$$\therefore \frac{dh}{dt} = \frac{dh}{dv} \times \frac{dv}{dt}$$

$$= \frac{3}{\pi h^2} \times 15$$

$$= \frac{45}{\pi h^2}$$

$$\text{When } h=4 \quad \frac{dh}{dt} = \frac{45}{16\pi} \text{ m/s} \quad [2]$$

Question 5:

$$a) f(x) = e^{x+2} \quad \left[\begin{array}{l} D: \text{all real } x \\ R: y > 0 \end{array} \right]$$

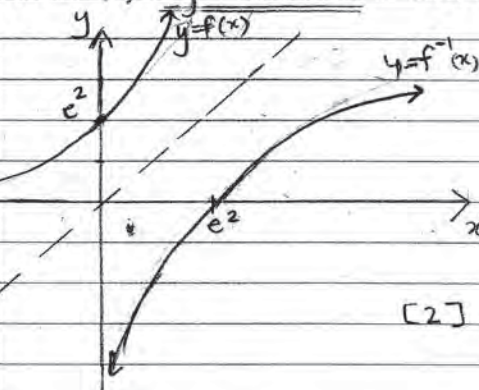
$$(i) y = e^{x+2}$$

$$x = e^{y+2}$$

$$\ln x = y+2$$

$$\ln x - 2 = y$$

$$\therefore f^{-1}(x): y = \ln x - 2 \quad [2]$$



[2]

$$b) \ddot{y} = -10$$

$$\dot{y} = -10t + c_1$$

$$\text{When } t=0 \quad \dot{y} = 10$$

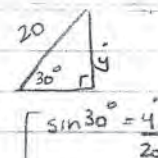
$$\therefore c_1 = 10$$

$$\dot{y} = -10t + 10$$

$$y = -5t^2 + 10t + c_2$$

$$t=0 \quad y=40 \therefore c_2 = 40$$

$$y = -5t^2 + 10t + 40 \quad [2]$$



$$\tan \theta = \frac{3}{\sqrt{3}}$$

$$\theta = 60^\circ$$

[3]

Question 6:

Step 1: Prove true for $n=1$

$$2^3 - 1 = 7 \text{ which is divisible by 7.}$$

Step 2: Assume true for $n=k$

$$\text{i.e. } 2^{3k} - 1 = 7M \text{ where } M \text{ is an integer}$$

$$2^{3k} = 7M + 1 \quad (1)$$

Now, prove true for $n=k+1$

$$2^{3(k+1)} - 1 = 2^{3k+3} - 1$$

$$= 2^{3k} \cdot 2^3 - 1$$

$$= (7M+1)8 - 1 \quad \text{from (1)}$$

$$= 56M + 8 - 1$$

$$= 56M + 7$$

$$= 7(8M+1)$$

which is divisible by 7.

Step 3: Since true for $n=1$, then

true for $n=1+1=2$, $n=3$ and so on for all integers $n \geq 1$.

[3]

$$b) \ddot{y} = -10$$

$$\dot{y} = -10t + c_1$$

$$\text{When } t=0 \quad \dot{y} = 10$$

$$\therefore c_1 = 10$$

$$\dot{y} = -10t + 10$$

$$y = -5t^2 + 10t + c_2$$

$$t=0 \quad y=40 \therefore c_2 = 40$$

$$y = -5t^2 + 10t + 40 \quad [2]$$

ii) Hits the lake when $y=0$

$$5t^2 - 10t - 40 = 0$$

$$t^2 - 2t - 8 = 0$$

$$(t-4)(t+2) = 0$$

$$t=4 \text{ or } -2 \text{ but } t > 0$$

$$\therefore t = 4 \text{ s}$$

$$\text{When } t=4 \quad x = 10\sqrt{3} \times 4$$

$$x = 40\sqrt{3} \text{ m} \quad [3]$$

$$(iii) x = 10\sqrt{3}t$$

$$\dot{x} = 10\sqrt{3}$$

$$\text{At } t=4 \quad \dot{y} = -40 + 10$$

$$\dot{y} = -30$$

$$\dot{v}^2 = \dot{x}^2 + \dot{y}^2$$

$$= (10\sqrt{3})^2 + (-30)^2$$

$$v = \sqrt{1200} \quad v > 0$$

$$\text{Speed} = 20\sqrt{3} \text{ m/s}$$

$$\tan \theta = \left| \frac{\dot{y}}{\dot{x}} \right| =$$

$$= \left| \frac{-30}{10\sqrt{3}} \right|$$

(b) $P(2ap, ap^2) \quad Q(2aq, aq^2)$

$$\begin{aligned} \text{(i)} \quad m_{PQ} &= \frac{aq^2 - ap^2}{2aq - 2ap} \\ &= \frac{a(q-p)(q+p)}{2a(q-p)} \\ &= \frac{p+q}{2} \end{aligned}$$

Eqn PQ:

$$y - ap^2 = \left(\frac{p+q}{2}\right)(x - 2ap)$$

$$y - ap^2 = \left(\frac{p+q}{2}\right)x - ap^2 - apq$$

$$y = \left(\frac{p+q}{2}\right)x - apq \quad [2]$$

(ii) since a focal chord, passes thro $(0, a)$

$$a = 0 - apq$$

$$\frac{a}{-a} = pq$$

$$pq = -1$$

[1]

(iii) $m_1 = \frac{p+q}{2} \quad m_2 = p$

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{p+q}{2} - p}{1 + \left(\frac{p+q}{2}\right)p} \right|$$

$$= \left| \frac{\frac{p+q-2p}{2}}{\frac{2+p^2+pq}{2}} \right|$$

$$= \left| \frac{p+q-2p}{2+p^2+pq} \right|$$

$$= \left| \frac{q-p}{p^2+pq+2} \right|$$

Now $pq = -1$
 $p = -\frac{1}{q}$

$$= \left| \frac{q + \frac{1}{q}}{\frac{1}{q^2} - 1 + 2} \right|$$

$$= \left| \frac{\frac{q^2+1}{q}}{\frac{1}{q^2}+1} \right|$$

$$= \left| \frac{\frac{q^2+1}{q}}{\frac{q^2+1}{q^2}} \right|$$

$$= |q|$$

$$\theta = \tan^{-1} |q| \quad [3]$$

(c) $\left(\frac{1}{3} + 2x\right)^{18}$

(i) $T_{k+1} = {}^{18}C_k a^{n-k} b^k$

$$= {}^{18}C_k \left(\frac{1}{3}\right)^{18-k} (2x)^k$$

$$= {}^{18}C_k \frac{1}{3^{18-k}} 2^k \cdot x^k$$

$$= {}^{18}C_k \frac{2^k}{3^{18-k}} x^k$$

[1]

(ii) $\frac{T_{k+1}}{T_k} = \frac{{}^{18}C_k \frac{2^k}{3^{18-k}}}{{}^{18}C_{k-1} \frac{2^{k-1}}{3^{18-k+1}}}$

$$= \frac{18!}{k!(18-k)!} \times \frac{(k-1)!(18-k+1)!}{18!} \times 2^{k-k+1} \times 3^{18-k+1-18+k}$$

$$= \frac{19-k}{k} \cdot 2 \cdot 3$$

$$= \frac{114-6k}{k}$$

Greatest coeff when $\frac{T_{k+1}}{T_k} \geq 1$

$$114-6k \geq k$$

$$114 \geq 7k$$

$$16\frac{2}{7} \geq k$$

$\therefore$ For $k=1, 2, 3, \dots, 16 \quad T_{k+1} > T_k$ and for $k=17, 18, \dots \quad T_{k+1} < T_k$

$\therefore$ Greatest coeff when $k=16$

Greatest Coefficient is ${}^{18}C_{16} \frac{2^{16}}{3^2} = \underline{\underline{114112}}$

[2]

$$i) y = 4 \cos 10t$$

$$\frac{dy}{dt} = -40 \sin 10t$$

$$\frac{d^2y}{dt^2} = -400 \cos 10t$$

$$\text{but } y = 4 \cos 10t$$

$$\therefore \frac{d^2y}{dt^2} = -100 (4 \cos 10t)$$

$$\frac{d^2y}{dt^2} = -100y \quad (n=10)$$

$\therefore$ since piston is moving in form $\ddot{y} = -ny$ then moving in SHM [2]

$$(ii) T = \frac{2\pi}{n}$$

$$= \frac{2\pi}{10}$$

$$= \frac{\pi}{5} \quad [1]$$

$$(iii) y = 3 \quad 3 = 4 \cos 10t$$

$$\frac{3}{4} = \cos 10t$$

$$\cos^{-1} \frac{3}{4} = 10t$$

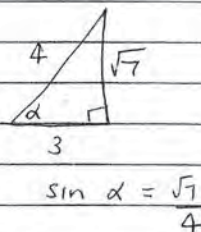
$$\frac{1}{10} \cos^{-1} \frac{3}{4} = t$$

$$\dot{y} = -40 \sin 10 \left(\frac{1}{10} \cos^{-1} \frac{3}{4} \right)$$

$$= -40 \sin \left(\cos^{-1} \frac{3}{4} \right)$$

$$= -40 \times \frac{\sqrt{7}}{4}$$

$$= -10\sqrt{7} \text{ cm/s} \quad [2]$$



$$\sin \alpha = \frac{\sqrt{7}}{4}$$

$$(iv) OB = OA + AB$$

$$\text{In } \triangle OAP, \cos \theta = \frac{OA}{4}$$

$$\sin \theta = \frac{AP}{4}$$

$$OA = 4 \cos \theta \quad (1)$$

$$AP = 4 \sin \theta$$

In $\triangle ABP$, by pythagoras theorem

$$BP^2 = AP^2 + AB^2$$

$$BP^2 = (4 \sin \theta)^2 + AB^2$$

$$BP^2 - 16 \sin^2 \theta = AB^2$$

$$\therefore AB = \sqrt{BP^2 - 16 \sin^2 \theta} \quad AB > 0 \quad (2) \quad [3]$$

$$(v) \left(\frac{d\theta}{dt} = 10 \right) \therefore OB = 4 \cos \theta + \sqrt{BP^2 - 16 \sin^2 \theta}$$

$$OB = 4 \cos \theta + (BP^2 - 16 \sin^2 \theta)^{\frac{1}{2}}$$

$$\frac{d(OB)}{d\theta} = -4 \sin \theta + \frac{1}{2} (BP^2 - 16 \sin^2 \theta)^{-\frac{1}{2}} \times -32 \sin \theta \cos \theta$$

$$= -4 \sin \theta - \frac{16 \sin \theta \cos \theta}{\sqrt{BP^2 - 16 \sin^2 \theta}}$$

$$= -4 \sin \theta - \frac{16 \sin \theta \cos \theta}{\sqrt{16 \left(\left(\frac{BP}{4} \right)^2 - \sin^2 \theta \right)}}$$

$$= -4 \sin \theta - \frac{16 \sin \theta \cos \theta}{4 \sqrt{\left(\frac{BP}{4} \right)^2 - \sin^2 \theta}}$$

$$= -4 \sin \theta - \frac{4 \sin \theta \cos \theta}{\sqrt{\left(\frac{BP}{4} \right)^2 - \sin^2 \theta}}$$

$$= -4 \sin \theta - \frac{4 \sin \theta \cos \theta}{\sqrt{\left(\frac{BP}{4} \right)^2 - \sin^2 \theta}}$$

$$= -4 \sin \theta \left[1 + \frac{\cos \theta}{\sqrt{\left(\frac{BP}{4} \right)^2 - \sin^2 \theta}} \right] \quad [3]$$

$$\text{Now given } \frac{d(OB)}{dt} = \frac{d(OB)}{d\theta} \times \frac{d\theta}{dt}$$

$$= -4 \sin \theta \left[1 + \frac{\cos \theta}{\sqrt{\left(\frac{BP}{4} \right)^2 - \sin^2 \theta}} \right] \times 10$$

$$= -40 \sin \theta \left[1 + \frac{\cos \theta}{\sqrt{\left(\frac{BP}{4}\right)^2 - \sin^2 \theta}} \right] \text{ which is required.}$$

(vi) when $\theta = 90^\circ$

$$\frac{d(OB)}{dt} = -40 \sin 90^\circ \left[1 + \frac{\cos 90^\circ}{\sqrt{\left(\frac{BP}{4}\right)^2 - \sin^2(90^\circ)}} \right]$$

$$\sin 90^\circ = 1 \quad \cos 90^\circ = 0$$

$$\therefore \frac{d(OB)}{dt} = -40 (1 + 0)$$

$$= \underline{\underline{-40 \text{ cm/s}}} \quad [1]$$